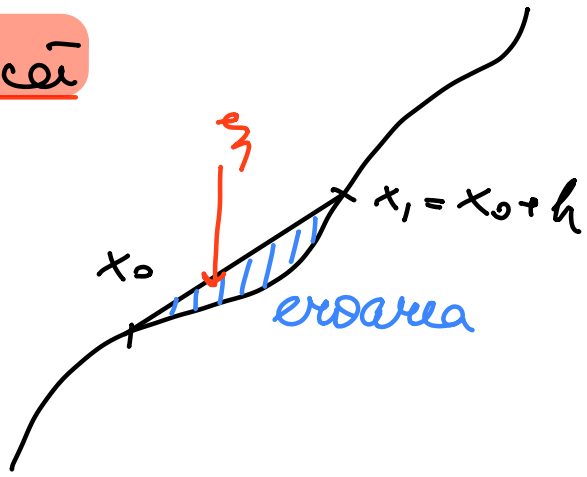
Derivare numerică

$f: \mathbb{R} \longrightarrow \mathbb{R} \in C^1$



$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$h << 1$ avem $f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h}$

$$f(x) = f(x_0)\frac{(x-x_1)}{x_0-x_1} + f(x_1)\frac{(x-x_0)}{x_1-x_0} + f''(\xi)\frac{(x-x_0)(x-x_1)}{2!}$$

P. Lagrange — $R_2(x)$

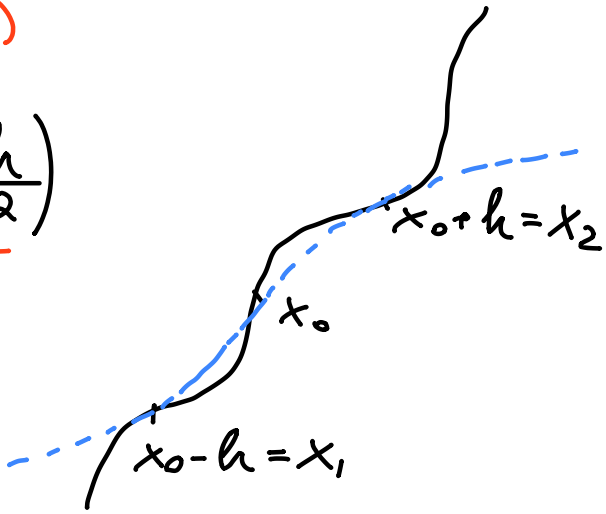
$$f'(x) = \frac{f(x_0)}{x_0-x_1} + \frac{f(x_1)}{x_1-x_0} + f''(\xi)\frac{2x-(x_0+x_1)}{2!}$$

$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + f''(\xi)\left(x - \frac{x_0+x_1}{2}\right)$$

$\mathcal{E}_2(x)$

$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + f''(\xi)\left(-\frac{h}{2}\right)$$

$O(h)$



$$f(x) = f(x_0)\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} +$$

$$+ f(x_1)\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + f(x_2)\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$+ f'''(\xi)\frac{(x-x_0)(x-x_1)(x-x_2)}{6}$$

$$f'(x_0) = \frac{1}{2h}\left(f(x_0+h) - f(x_0-h)\right) + \underbrace{\frac{1}{6}f'''(\xi)O(h^2)}_{O(h^2)}$$

Integrare numerică

$$I[f] = \int_a^b f(x)\,dx$$

$$I_N[f] = \sum_{i=1}^{N} A_{iN} f(x_{iN}) \quad \text{Cuadratură}$$

Quadratură = mult. de pct. $\{A_i, x_i\}_{i \in \overline{1,N}}$

a.î. $\int_a^b f(x)\,dx = \sum_{i=0}^{N} A_i f(x_i)$ → valabilă pt. $\forall$ polinom cu grad $\leq N$

$\Rightarrow$ PUTEREA METODEI E $(N+1)$

Th. Condiții necesare și suficiente

$I_N[f]$ să conveargă slab la $I[f]$

$I_N[f] \xrightarrow{\text{slab}} I[f]$

este dată de

a) $\exists M > 0$ a.î. $\sum_{i=1}^{N} |A_{iN}| < M$

b) $\lim_{N \to \infty} I_N[x^k] = \int_a^b x^k\,dx \quad k = 0, 1, 2, \ldots$

$k=0$

$$\begin{cases} \sum_{i=1}^{N} A_{iN} = \int_a^b dx \\ \sum_{i=1}^{N} A_{iN} x_{iN} = \int_a^b x\,dx \\ \dots\dots \\ \sum_{i=1}^{N} A_{iN} x_{iN}^{k} = \int_a^b x^k dx \end{cases} \qquad k = 0 \dots 2N-1$$

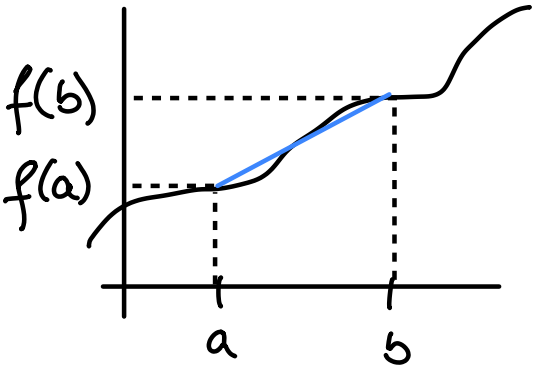
$$x_{iN} = a + ih \quad \text{cu} \quad h = \frac{b-a}{N} \qquad i = 0:N$$

atunci $k = 0 \dots N-1$

(1) Newton - Cotes

$$\int_a^b f(x)w(x)\,dx = \sum_{i=1}^{N} A_{iN} f(x_{iN}) + R_N$$

(a) Metoda trapezului



$$\int_a^b f(x)dx \simeq \frac{f(a)+f(b)}{2}(b-a) + \mathcal{E}_{a,b}$$

$$\int_a^{a+h} f(x)dx \simeq \frac{f(a)+f(a+h)}{2}\cdot h + \mathcal{E}_h \longrightarrow O(h^3)$$

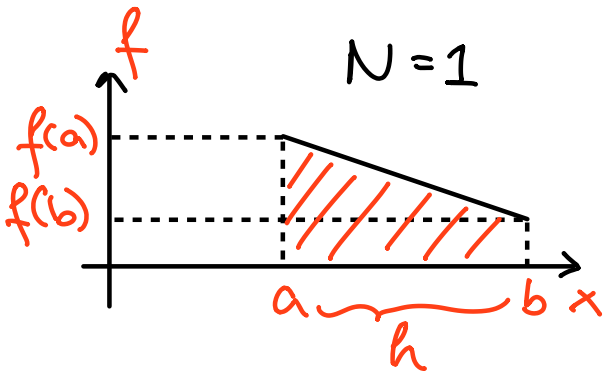
(b) Metoda compusă a trapezului

→ luăm $x_{iN} = a + ih$ cu $h = \frac{b-a}{N}$ $\quad i = 0:N$

→ $f(x) = P_{n-1}(x) = \sum_{i=1}^{N} f(x_{iN}) l_i(x)$

Lagrange

$$A_{iN} = \int_a^b l_i(x) \cdot w(x)\,dx$$



$$\int_a^b f(x) w(x) dx = \frac{h}{2}\left[f(a) + f(b)\right]$$

$[a,b] \longrightarrow x_i = a + ih$

$$\int_{x_i}^{x_{i+1}} f(x) w(x) dx = \frac{h}{2}\left[f(x_i) + f(x_{i+1})\right]$$

$$\int_a^b f(x) w(x) dx = \int_{x_0}^{x_1} f(x) w(x)\,dx + \dots + \int_{x_{N-1}}^{x_N} f(x) w(x)\,dx$$

$$= \frac{f(x_0)+f(x_1)}{2} \cdot h + \frac{f(x_1)+f(x_2)}{2} \cdot h + \dots + \frac{f(x_{N-1})+f(x_N)}{2} + \mathcal{E}_h$$

$$= \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{N-1}) + f(x_N)\right] + \mathcal{E}_h$$

$$\int_a^b = \int_a^{x_1} + \int_{x_1}^{x_2} + \int_{x_2}^{x_3} + \dots + \int_{x_{n-1}}^{b}$$

$$\int_a^b f(x)w(x)\,dx = \frac{h}{2}\left[f(a)+f(b)+2\sum_{i=1}^{N-1} f(a+ih)\right]$$

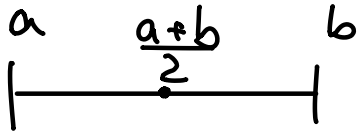
formula trapezelor

$$h = \frac{b-a}{N}$$

Eroare: $R_N = -\frac{h^3 f''(\xi)}{12}\cdot N$

(b) **Simpson**

pt. $N=2$ avem

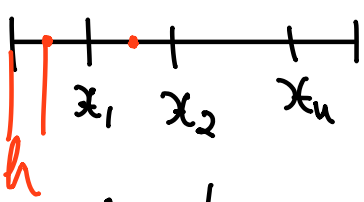


polinoamele Lagrange $l_0 = \left(x-\frac{a+b}{2}\right)(x-b)$

$l_1 = (x-a)(x-b)$

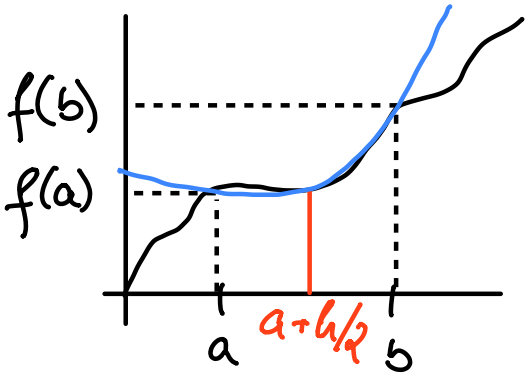
$l_2 = (x-a)\left(x-\frac{a+b}{2}\right)$

prin integrare $\int = \frac{h}{3}\left[f(a)+f(b)+4f\left(\frac{a+b}{2}\right)\right]$ $\quad h = \frac{b-a}{N}$



→ renumerotez punctele ca

$h = \frac{b-a}{2N}$ $\quad x_i = a+ih$ $\quad i = 0:2N$



$$\int_a^b f(x)\,dx = \int_{x_0}^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx + \dots + \int_{x_{N-1}}^{x_N} f(x)\,dx$$

$$y = \frac{h}{3}\left[f(a) + f(b) + 4\sum_{i=1}^{N} f(x_{2i-1}) + 2\sum_{i=1}^{N-1} f(x_{2i})\right]$$

formula Simpson

Eroare: $E = -\frac{h^5 f^{(4)}(\xi)}{90} \cdot N$

(2) Cuadraturi gaussiene

→ extinderea gradului de valabilitate

$$\overline{\Pi}_N = (x - x_{1N}) \dots (x - x_{NN})$$

$$P_{N+q-1}(x) = \overline{\Pi}_N(x) \cdot Q_{q-1}(x) + R_{N-1}(x)$$

$$\int_a^b P_{N+q-1}(x) \cdot w(x)\,dx = \sum_{i=1}^{N} A_{iN} P_{N+q-1}(x_{iN})$$

$$P_{N+q-1}(x_{iN}) = R_{N-1}(x_{iN})$$

$$\int_a^b P_{N+q-1}(x)\cdot\omega(x)\,dx = \int_a^b \Pi_N(x)\cdot Q_{q-1}(x)\omega(x)\,dx + \int_a^b R_{N-1}(x)\,\omega(x)\,dx$$

$$\int_a^b P_{N+q-1}(x)\cdot\omega(x)\,dx = \sum_{i=1}^{N} A_{iN} R_{N-1}(x_{iN})$$

$$\int_a^b \Pi_N(x)\cdot Q_{q-1}(x)\omega(x)\,dx = 0$$

$$\Rightarrow q = N$$

$\Pi_N(x)$ polinom ortogonal relativ la $[a,b]$ și funcția pondere $\omega(x)$

gradul de valabilitate $2N-1$

suportul integrării x_{iN} - rădăcinile lui $\Pi_N(x)$

A_{iN} se determină relativ la $[a,b]$ și $\omega(x)$ $\left(\Pi_N(x)\right)$

Teoremă

Dacă x_i sunt rădăcinile unui polinom ortogonal, atunci puterea metodei este $2N+1$

$$\int_{-1}^{1} P\left(\frac{x-\frac{a+b}{2}}{\frac{a-b}{2}}\right)dx = \int_a^b P(x)\,dx$$

Polinoame ortogonale

Cebâșev	$[-1;1]$	$w(x)=\frac{1}{\sqrt{1-x}}$
Legendre	$[-1;1]$	$w(x)=1$
Laguerre	$(0,\infty)$	$w(x)=e^{-x}$
Hermite	$(-\infty;\infty)$	$w(x)=e^{-x^2}$

Problemă examen

Determinați forma de integrare pentru

$$\int_0^1 f(x)\,dx = a_0 f(0) + a_1 f(1) + b_0 f(x_0) + b_1 f(x_1)$$

a.î. funcția să aibă gradul maxim posibil

gr. max = 7

$$\overline{T}_4(x) = x(x-1)(x-x_0)(x-x_1) = (x^2-x)(x^2-Sx+P)$$

$$\int_0^1 \overline{T}_4(x)\cdot x^0\,dx = 0 \Rightarrow \frac{1}{5}-\frac{S}{4}+\frac{P}{3}-\frac{1}{4}+\frac{S}{3}-\frac{P}{2}=0$$

$$\int_0^1 \overline{T}_4(x)\cdot x^1\,dx = 0 \Rightarrow \frac{1}{6}-\frac{S}{5}+\frac{P}{4}-\frac{1}{5}+\frac{S}{4}-\frac{P}{3}=0$$

$$\Rightarrow \begin{cases} S\left(\frac{1}{3}-\frac{1}{4}\right)+P\left(\frac{1}{3}-\frac{1}{2}\right)=\frac{1}{4}-\frac{1}{5} \\ S\left(\frac{1}{4}-\frac{1}{5}\right)+P\left(\frac{1}{4}-\frac{1}{3}\right)=\frac{1}{5}-\frac{1}{6} \end{cases} \Rightarrow \begin{cases} \frac{S}{12}-\frac{P}{6}=\frac{1}{20} \\ \frac{S}{20}-\frac{P}{12}=\frac{1}{30} \end{cases}$$

$$S\left(\frac{1}{20}-\frac{1}{24}\right)=\frac{1}{30}-\frac{1}{40} \Rightarrow S\cdot\frac{4}{20\cdot 24}=\frac{10}{30\cdot 40}$$

$$S=1$$

$$P=\left(\frac{1}{12}-\frac{1}{20}\right)\cdot 6=\frac{1}{2}-\frac{3}{10}=\frac{2}{10}=\frac{1}{5}$$

$$x^2-x+\frac{1}{5}=0 \qquad x_{0,1}=\frac{1\pm\frac{1}{\sqrt{5}}}{2}=\frac{5\mp\sqrt{5}}{10}$$

$$\int_0^1 1\,dx = a_0+a_1+b_0+b_1=1$$

$$\int_0^1 x\,dx = \qquad a_1+b_0\frac{5-\sqrt{5}}{10}+b_1\frac{5+\sqrt{5}}{10}=\frac{1}{2}$$

$$\int_0^1 x^2\,dx = \qquad a_1+b_0\frac{3-\sqrt{5}}{10}+b_1\frac{3+\sqrt{5}}{10}=\frac{1}{3}$$

$$\begin{cases} a_0+a_1+b_0+b_1=1 \\ a_1+\frac{b_0}{2}+\frac{b_1}{2}=\frac{1}{2} \\ b_0=b_1 \\ a_1+\frac{3b_0}{10}+\frac{3b_1}{10}=\frac{1}{3} \end{cases}$$

$$\begin{cases} a_0+a_1+2b_0=1 \\ a_1+b_0=\frac{1}{2} \\ a_1+\frac{3b_0}{5}=\frac{1}{3} \end{cases} \Rightarrow \frac{2}{5}b_0=\frac{1}{6} \Rightarrow b_0=\frac{5}{12}$$

$$b_1=\frac{5}{12}$$

$$a_1 = \frac{1}{2} - \frac{5}{12} = \frac{1}{12}$$

$$a_0 = 1 - \frac{1}{12} - \frac{10}{12} = \frac{1}{12}$$

$$\int_0^1 f(x)\,dx = \frac{1}{12}\left[f(0) + f(1) + 5f\left(\frac{5-\sqrt{5}}{10}\right) + 5f\left(\frac{5+\sqrt{5}}{10}\right)\right]$$

$$= \ln(1+x^2)$$

Problemă examen

Determinați forma de integrare pentru

$$\int_{-1}^{1} f(x)\,dx = k\left(f(x_1) + f(x_2) + f(x_3)\right)$$

a.î. funcția să aibă gradul maxim posibil

grad max $= 2 \cdot 3 - 1 = 6 - 1 = 5$

$$\int_{-1}^{1} 1\,dx = k(1+1+1) \Rightarrow x\Big|_{-1}^{1} = 3k \Rightarrow$$

$$\Rightarrow 2 = 3k \Rightarrow k = \frac{2}{3}$$

$$\Pi_3(x) = (x-x_1)(x-x_2)(x-x_3) =$$

$$= (x-x_1)(x^2 - Sx + P)$$

$$\int_{-1}^{1} \Pi_3(x) \cdot x^0\,dx = 0 \Rightarrow \frac{1}{4} - \frac{S}{3} + \frac{P}{2} - \frac{x_1}{3} + \frac{Sx_1}{2} - \frac{Px_1}{1} -$$

$$-\left(\frac{1}{4} + \frac{S}{3} + \frac{P}{2} + \frac{x_1}{3} + \frac{Sx_1}{2} + \frac{Px_1}{1}\right) = 0$$

$$\int_{-1}^{1} \pi_3(x)\cdot x^1\,dx = 0 \Rightarrow \frac{1}{5} - \frac{S}{4} + \frac{P}{3} - \frac{x_1}{4} + \frac{Sx_1}{3} - \frac{P}{2} -$$

$$-\left(\frac{-1}{5} - \frac{S}{4} - \frac{P}{3} - \frac{x_1}{4} - \frac{Sx_1}{3} - \frac{P}{2}\right) = 0$$

$$\int_{-1}^{1} \pi_3(x)\cdot x^2\,dx = 0 \Rightarrow \frac{1}{6} - \frac{S}{5} + \frac{P}{4} - \frac{x_1}{5} + \frac{Sx_1}{4} - \frac{P}{3}$$

$$-\left(\frac{1}{6} + \frac{S}{5} + \frac{P}{4} + \frac{x_1}{5} + \frac{Sx_1}{4} + \frac{P}{3}\right) = 0$$

$$\begin{cases} -\frac{2S}{3} - \frac{2x_1}{3} - Px_1 = 0 \mid \cdot 3 \\ \frac{2}{5} + \frac{2P}{3} + \frac{2Sx_1}{3} = 0 \mid \cdot \frac{15}{2} \\ -\frac{2S}{5} - \frac{2x_1}{5} - \frac{2P}{3} = 0 \mid \cdot \frac{15}{2} \end{cases}$$

$$\Rightarrow \begin{cases} -2S - 2x_1 - Px_1 = 0 \\ 3 + 5P + 5Sx_1 = 0 \\ -3S - 3x_1 - 5P = 0 \end{cases}$$